

Unit 1

Design Thinking as Mindset

3 Slibraries · 15 framework sheets

SLIBRARY 1 · DT Foundations

5 sheets

- | | |
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| 1 Tim Brown's DT Triple | 4 IDEO 3 Lenses of HCD |
| 2 Double Diamond | 5 Brenner's DT Holistic Model |
| 3 Stanford d.school 5-Stage Process | |

SLIBRARY 2 · DT Mindsets

5 sheets

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|--------------------------------|--|
| 1 Liedtka's Why DT Works | 4 Buchanan's Wicked Problems |
| 2 Dosi's DT Mindset Measure | 5 Bason & Austin: Right Way to Lead DT |
| 3 Kelley's Creative Confidence | |

SLIBRARY 3 · DT Process & Tools

5 sheets

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|------------------------|----------------------|
| 1 Empathy Map | 4 5 Whys |
| 2 Persona Canvas | 5 How Might We (HMW) |
| 3 Customer Journey Map | |

Tim Brown's DT Triple

SLIBRARY 1 · SHEET 1 / 5

Three overlapping spaces, not sequential steps

PROFESSIONAL DEFINITION

Tim Brown's foundational account of design thinking organises the work into three overlapping spaces — Inspiration, Ideation and Implementation — rather than a linear sequence of steps. Inspiration surfaces the problem through context and empathy; Ideation generates and prototypes solutions; Implementation brings the chosen solution to life. Projects loop back through the spaces as learning accumulates, and any space can be re-entered when new evidence arrives (Brown, 2008; Brown and Wyatt, 2010).

WHO DEVELOPED IT

Articulated by Tim Brown, then CEO of IDEO, in his 2008 *Harvard Business Review* article *Design Thinking*, drawing on three decades of IDEO consulting practice and David Kelley's earlier Stanford and IDEO methodology.

WHY IT WAS DEVELOPED

Brown wanted to liberate design from a phase-gate process metaphor — which mis-framed it as a final "styling" step — and re-position it as a mindset for innovation. Treating the work as overlapping spaces makes iteration a feature, not a bug, of doing it well.

HOW IT IS USED

Teams move fluidly between the three spaces, with frequent looping. Each space uses different tools (ethnographic field-work, brainstorming, prototyping) but shares one rule: everything is provisional until evidence arrives.

WHEN IT IS USED

Use early in any innovation project where the problem is not yet well-defined, when re-framing matters more than execution speed, and whenever a team is tempted to skip from Inspiration straight to Implementation.

BY WHOM IT IS USED

IDEO consultants, in-house innovation teams, MBA design-thinking courses, and senior managers leading multi-disciplinary teams in product, service and policy contexts.

FIGURE 1 · BROWN'S THREE SPACES

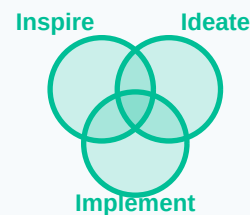


Figure 1. The three spaces overlap; projects loop back as learning accumulates. Adapted from Brown (2008).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Herbert Simon (1969) framed design as iterative problem-solving and the Stanford d.school's 5-stage process (c. 2005) named distinct phases, Brown's contribution was to frame the work as *overlapping* rather than *stepwise*. The triple-space mental model gave non-designers permission to revisit earlier work — the move that opened design thinking to general management.

RELATED FRAMEWORKS IN THIS COURSE

IDEO 3 Lenses of HCD · Slibrary 1

Stanford d.school 5-Stage Process · Slibrary 1

Double Diamond · Slibrary 1

Brenner's DT Holistic Model · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Brown, T. (2008) 'Design Thinking', *Harvard Business Review*, 86(6), pp. 84–92.

Brown, T. and Wyatt, J. (2010) 'Design Thinking for Social Innovation', *Stanford Social Innovation Review*, 8(1), pp. 30–35.

Brown, T. (2009) *Change by Design*. New York: HarperBusiness.

Kelley, T. and Littman, J. (2001) *The Art of Innovation*. New York: Doubleday.

Double Diamond

SLIBRARY 1 · SHEET 2 / 5

Discover → Define → Develop → Deliver

PROFESSIONAL DEFINITION

The Double Diamond is a four-stage process model — Discover, Define, Develop, Deliver — visualised as two diamonds representing alternating divergence and convergence. The first diamond explores the problem space; the second diamond explores the solution space. Each diamond opens widely (divergence) before narrowing to a chosen direction (convergence), making the discipline of generating options before deciding visually unmissable (Design Council, 2005, 2019).

WHO DEVELOPED IT

Created by the British Design Council in 2005, refined and re-issued in 2019 as the *Framework for Innovation*. The model synthesises decades of design and engineering process literature into a single, teachable shape.

WHY IT WAS DEVELOPED

Existing process diagrams either compressed problem-finding into a single step or merged it with solving. The Double Diamond made the discipline of staying in the problem space — before any solving begins — visually unmissable, addressing the most common failure mode of beginner designers.

HOW IT IS USED

Teams use divergent methods (research, brainstorming) on the way out, and convergent methods (synthesis, prioritisation) on the way back. The narrow points are decision moments, named explicitly.

WHEN IT IS USED

Use to teach iterative problem-framing, to plan project timelines that allocate real time to discovery, and to communicate progress to stakeholders who expect linear plans.

BY WHOM IT IS USED

UK government service designers, Design Council practitioners, undergraduate and MBA design courses, and any team needing a shared visual language for the divergence-convergence rhythm.

FIGURE 1 · FOUR PHASES

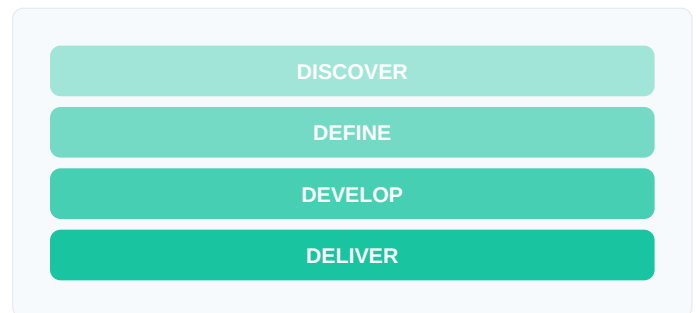


Figure 1. The four phases alternate divergent and convergent thinking; the two diamonds separate problem from solution.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Herbert Simon's (1969) generic design loop and Tim Brown's (2008) three-space model, the Double Diamond's distinctive contribution is the *two diamonds* — separating problem definition from solution development. This decoupling, anticipated by Christopher Jones (1970), made it the canonical visual for service-design education.

RELATED FRAMEWORKS IN THIS COURSE

Tim Brown's DT Triple · Slibrary 1

Stanford d.school 5-Stage Process · Slibrary 1

Brenner's DT Holistic Model · Slibrary 1

How Might We (HMW) · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Design Council (2005) *The Design Process: What is the Double Diamond?*. London: Design Council.

Design Council (2019) *Framework for Innovation: Design Council's Evolved Double Diamond*. London: Design Council.

Tschimmel, K. (2012) 'Design Thinking as an Effective Toolkit for Innovation', in *ISPIM Conference Proceedings*, pp. 1–20.

Jones, J.C. (1970) *Design Methods: Seeds of Human Futures*. London: Wiley.

Stanford d.school 5-Stage Process

SLIBRARY 1 · SHEET 3 / 5

Empathize · Define · Ideate · Prototype · Test

PROFESSIONAL DEFINITION

The Stanford d.school's five-stage process — Empathize, Define, Ideate, Prototype, Test — is the most widely taught operationalisation of design thinking globally. The stages are non-linear and frequently looped: insights from Test feed back into Define, prototypes generate empathy that re-opens the problem. The model functions as a shared vocabulary for cross-disciplinary teams rather than a strict sequence (Plattner, 2010; Both and Baggereor, 2013).

WHO DEVELOPED IT

Developed at Stanford's Hasso Plattner Institute of Design (the d.school), founded in 2005 by David Kelley with seed funding from Hasso Plattner. The five-stage formulation was codified in the d.school's *Bootcamp Bootleg* (2010) and disseminated through executive and university courses.

WHY IT WAS DEVELOPED

The d.school needed a teachable scaffold that could be picked up by engineering, business, medicine and humanities students within days. A five-stage model — concrete enough to remember, generic enough to apply anywhere — solved the pedagogical problem.

HOW IT IS USED

Teams progress through each stage using stage-specific tools (empathy interviews, POV statements, brainstorming, low-fi prototypes, user testing) but loop back as learning emerges.

WHEN IT IS USED

Use as the on-ramp for novices learning design thinking, as a structured engagement template for client workshops, and as a checklist when a team is stuck or has skipped a step.

BY WHOM IT IS USED

Stanford d.school students, executive education participants worldwide, school-level K–12 design programmes, and corporate innovation teams adopting design thinking for the first time.

FIGURE 1 · FIVE STAGES

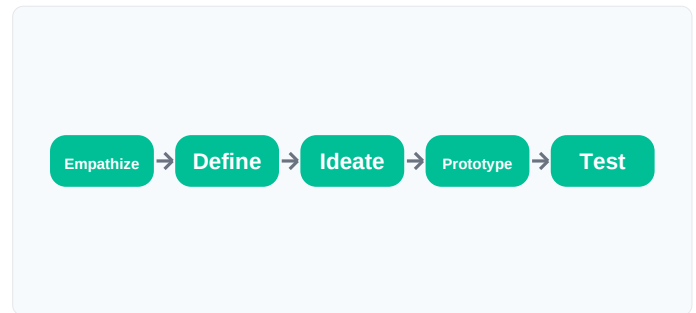


Figure 1. The five stages are taught as sequential but practised as iterative; loop-backs are the rule, not the exception.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Tim Brown's (2008) Triple Loop kept the spaces deliberately fuzzy and the Double Diamond (Design Council, 2005) emphasised divergence/convergence, the d.school's contribution was a memorable five-step heuristic. Building on Bruce Archer's (1965) systematic methodology, it made the work concretely teachable.

RELATED FRAMEWORKS IN THIS COURSE

Tim Brown's DT Triple · Slibrary 1

Double Diamond · Slibrary 1

Empathy Map · Slibrary 3

How Might We (HMW) · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Plattner, H. (2010) *Bootcamp Bootleg*. Stanford, CA: Hasso Plattner Institute of Design.

Both, T. and Baggereor, D. (2013) *d.school Design Thinking Bootleg*. Stanford, CA: d.school.

Carleton, T. and Leifer, L. (2009) 'Stanford's ME310 Course as an Evolution of Engineering Design', *Proceedings of the 19th CIRP Design Conference*, pp. 547–554.

Kelley, D. and Kelley, T. (2013) *Creative Confidence*. New York: Crown Business.

IDEO 3 Lenses of HCD

SLIBRARY 1 · SHEET 4 / 5

Desirability × Feasibility × Viability — the DVF triangle

PROFESSIONAL DEFINITION

IDEO's three-lens model frames every design decision against three independent tests: Desirability (do users want it?), Feasibility (can it be built and operated?) and Viability (can the business sustain it?). A successful innovation lies at the intersection of all three. Failures usually trace to having satisfied only one or two — for example, technically feasible products that no one wants, or desirable products that cannot be profitably delivered (IDEO, 2015; Brown, 2009).

WHO DEVELOPED IT

Codified by IDEO under Tim Brown and David Kelley in the late 1990s, drawing on earlier marketing concepts of consumer-needs research and engineering's build-test culture. Most cleanly articulated in IDEO's *Field Guide to Human-Centered Design* (2015).

WHY IT WAS DEVELOPED

IDEO's consultants needed a shared evaluation lens that designers, engineers and business strategists could all apply to the same idea. Without it, teams routinely optimised one dimension and shipped products that failed because of the other two.

HOW IT IS USED

At every decision point — concept selection, prototype review, go/no-go gates — the team scores each option against all three lenses. Options strong on all three advance; options weak on one are redesigned, not deleted.

WHEN IT IS USED

Use during ideation triage, business-model design, investor due diligence, and as a kill-switch criterion in stage-gate processes. Especially powerful when the team's natural bias favours one lens.

BY WHOM IT IS USED

Cross-functional product teams, MBA students evaluating ventures, IDEO and IDEO-trained consultants, and any team building business cases for innovation portfolios.

FIGURE 1 · DVF INTERSECTION

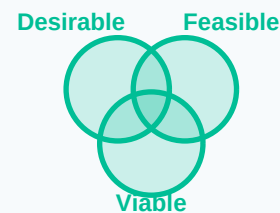


Figure 1. The innovation "sweet spot" lies where all three lenses overlap; satisfying only two is a known failure mode.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Edward de Bono's (1967) lateral-thinking emphasis on novel options and Geoffrey Moore's (1991) market-fit lens, the DVF model's contribution is its insistence that all three tests are simultaneous — not sequential. Bostelaar's MBA301 integration extends DVF as the evidence test in business-model and pitch design.

RELATED FRAMEWORKS IN THIS COURSE

Tim Brown's DT Triple · Slibrary 1

BMC 9 Blocks · Slibrary 9

Lean Canvas (9 Blocks) · Slibrary 14

Brenner's DT Holistic Model · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

IDEO.org (2015) *The Field Guide to Human-Centered Design*. San Francisco: IDEO.org.

Brown, T. (2009) *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation*. New York: HarperBusiness.

Kelley, T. and Littman, J. (2001) *The Art of Innovation: Lessons in Creativity from IDEO*. New York: Doubleday.

Moore, G.A. (1991) *Crossing the Chasm*. New York: HarperBusiness.

Brenner's DT Holistic Model

SLIBRARY 1 · SHEET 5 / 5

Design Thinking as Mindset · Process · Toolbox (Brenner, 2016)

PROFESSIONAL DEFINITION

Walter Brenner and colleagues' holistic model frames design thinking as the integration of three layers: a *mindset* (collaborative, user-centred, optimistic), a *process* (iterative, divergent-then-convergent, bias toward prototyping) and a *toolbox* (the field-instruments — empathy maps, personas, journey maps and so on). The model insists that any one layer alone is insufficient: tools without mindset become bureaucratic, and mindset without process becomes anecdotal (Brenner et al., 2016).

WHO DEVELOPED IT

Walter Brenner and colleagues at the University of St. Gallen and Hasso Plattner Institute, published in 2016 in *Design Thinking for Innovation*. Builds on Tschepe's (2017) earlier articulation of the same three-layer scheme.

WHY IT WAS DEVELOPED

Practitioners had spent a decade arguing whether design thinking was a process, a mindset, or a set of tools. Brenner's contribution was to dissolve the argument by treating all three as equally constitutive — and showing that adoption fails when only one is imported.

HOW IT IS USED

Teams deliberately develop all three layers in parallel — building mindset through coached practice, process through repeated project cycles, and toolbox fluency through workshop training.

WHEN IT IS USED

Use when assessing organisational readiness to adopt DT, when diagnosing why a DT initiative has stalled, and when designing internal training curricula for innovation teams.

BY WHOM IT IS USED

HR and L&D; leads designing DT capabilities, academic researchers studying DT adoption, consultancies training corporate teams, and reflective practitioners diagnosing their own practice gaps.

FIGURE 1 · THREE-LAYER MODEL

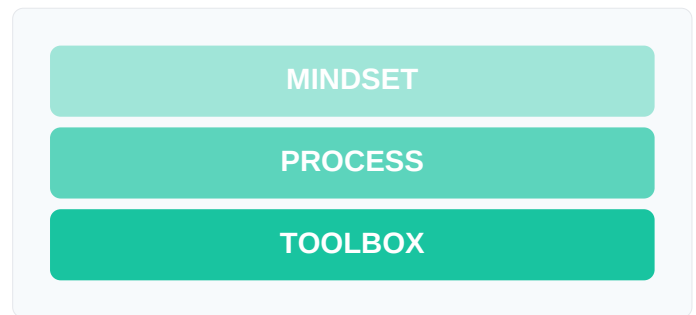


Figure 1. Adoption succeeds only when all three layers are imported together; isolated layers reliably fail.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Tim Brown (2008) emphasised the mindset, the Stanford d.school (2010) emphasised the process, and Tom Kelley (2001) emphasised the toolbox, Brenner's contribution is the *integration* argument — that all three must be cultivated simultaneously, drawing on Schein's (1985) culture model in adapting it to innovation practice.

RELATED FRAMEWORKS IN THIS COURSE

Tim Brown's DT Triple · Slibrary 1

Stanford d.school 5-Stage Process · Slibrary 1

Liedtka's Why DT Works · Slibrary 2

Dosi's DT Mindset Measure · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Brenner, W., Uebernickel, F. and Abrell, T. (2016) 'Design Thinking as Mindset, Process, and Toolbox', in Brenner, W. and Uebernickel, F. (eds) *Design Thinking for Innovation*. Cham: Springer, pp. 3–21.

Tschepe, S. (2017) 'Mindset, Process, Method: A Comprehensive Descriptive Model for Design Thinking', *HPI Working Papers*.

Plattner, H., Meinel, C. and Leifer, L. (eds) (2011) *Design Thinking: Understand – Improve – Apply*. Berlin: Springer.

Schein, E.H. (1985) *Organizational Culture and Leadership*. San Francisco: Jossey-Bass.

Liedtka's Why DT Works

SLIBRARY 2 · SHEET 1 / 5

Four mechanisms of effective design thinking

PROFESSIONAL DEFINITION

Jeanne Liedtka's research-based account argues that design thinking works because it counteracts four well-documented cognitive biases. Empathic immersion counteracts projection; structured ideation counteracts hypothesis-confirmation; prototyping and live-data experimentation counteract over-confidence; and the discipline of diversity counteracts in-group thinking. Design thinking succeeds, in Liedtka's argument, not because designers are special but because its tools impose better epistemic hygiene on ordinary teams (Liedtka, 2018).

WHO DEVELOPED IT

Jeanne Liedtka, professor at the University of Virginia Darden School. Her 2018 *Harvard Business Review* article *Why Design Thinking Works* distilled twenty years of empirical research across 50 organisations.

WHY IT WAS DEVELOPED

Despite design thinking's spread, sceptics challenged whether it produced systematically better outcomes than other innovation methods. Liedtka built an evidence-based answer rooted in cognitive psychology, not designer mythology — explaining the mechanism, not just claiming the result.

HOW IT IS USED

Teams audit their innovation practice against the four mechanisms — checking whether they are doing real empathy, structured ideation, live experimentation, and diverse-team facilitation, rather than performing them theatrically.

WHEN IT IS USED

Use when justifying design thinking to sceptical executives, when designing innovation training, and when diagnosing why a DT initiative has failed to deliver expected results.

BY WHOM IT IS USED

Innovation researchers, MBA professors, executive coaches, and corporate innovation leads making the business case for DT investment.

FIGURE 1 · FOUR BIAS COUNTER-MECHANISMS

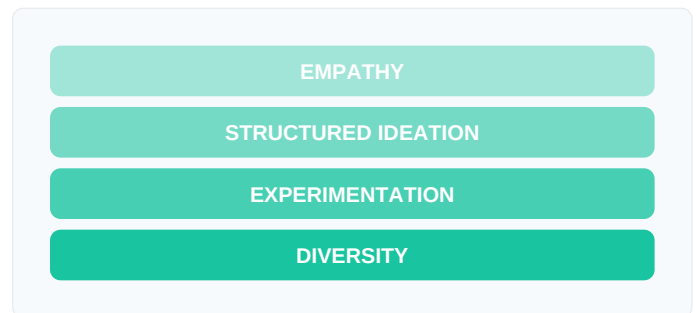


Figure 1. Each mechanism counteracts a specific cognitive bias documented in the behavioural-economics literature.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where earlier accounts (Brown, 2008; Plattner, 2010) described *what* design thinkers do, Liedtka's contribution was an evidence-based account of *why it works*, rooted in Kahneman's (2011) bias literature. The shift from descriptive to causal-mechanism explanation gave DT serious academic standing.

RELATED FRAMEWORKS IN THIS COURSE

Brenner's DT Holistic Model · Slibrary 1

Dosi's DT Mindset Measure · Slibrary 2

Kelley's Creative Confidence · Slibrary 2

Bason & Austin: Right Way to Lead DT · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

- Liedtka, J. (2018) 'Why Design Thinking Works', *Harvard Business Review*, 96(5), pp. 72–79.
 Liedtka, J. and Ogilvie, T. (2011) *Designing for Growth: A Design Thinking Tool Kit for Managers*. New York: Columbia Business School Publishing.
 Kahneman, D. (2011) *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux.
 Liedtka, J. (2015) 'Perspective: Linking Design Thinking with Innovation Outcomes', *Journal of Product Innovation Management*, 32(6), pp. 925–938.

Dosi's DT Mindset Measure

SLIBRARY 2 · SHEET 2 / 5

10-dimension measurement of the designerly mindset

PROFESSIONAL DEFINITION

Clio Dosi and colleagues developed the first psychometrically validated scale for measuring an individual's design-thinking mindset — a 10-dimension instrument covering tolerance for ambiguity, empathy, optimism, experimentation, comfort with risk, abductive reasoning, integrative thinking, embracing diversity, learning from failure, and bias for action. Each dimension is measured by validated Likert items, allowing organisations to baseline DT capability and track development over time (Dosi, Rosati and Vignoli, 2018).

WHO DEVELOPED IT

Clio Dosi, Francesco Rosati and Matteo Vignoli at the University of Bologna and Politecnico di Milano. Published in 2018 at the Design Research Society conference following multi-stage scale development with over 600 respondents.

WHY IT WAS DEVELOPED

Brenner's (2016) holistic model identified mindset as one of three layers but offered no way to measure it. Without measurement, organisations could not show progress, target weak dimensions, or compare candidates. Dosi's instrument closed the measurement gap and made mindset an empirical, not anecdotal, construct.

HOW IT IS USED

Administer the validated questionnaire (typically 40–50 items) before and after training programmes. Aggregate scores show capability, individual profiles target coaching, and dimension-level analysis informs curriculum design.

WHEN IT IS USED

Use to baseline cohorts before DT training, to evaluate ROI on capability-building investments, to screen candidates for innovation roles, and to track team-level capability over time.

BY WHOM IT IS USED

L&D; and HR analytics teams, academic researchers studying DT efficacy, executive coaches, and innovation programme directors needing measurable outcomes.

FIGURE 1 · TEN MINDSET DIMENSIONS



Figure 1. Ten dimensions, each independently scored. Profiles diagnose specific developmental needs.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Liedtka (2018) argued *that* mindset matters and Brenner et al. (2016) named it as a layer, Dosi's contribution is a validated measurement instrument. Building on Hambrick's (1981) executive cognition tradition and Spreitzer's (1995) empowerment scale, it allows DT mindset to be empirically tracked.

RELATED FRAMEWORKS IN THIS COURSE

Liedtka's Why DT Works · Slibrary 2

Kelley's Creative Confidence · Slibrary 2

Brenner's DT Holistic Model · Slibrary 1

Growth vs Fixed Mindset · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

- Dosi, C., Rosati, F. and Vignoli, M. (2018) 'Measuring Design Thinking Mindset', in *Proceedings of the DRS 2018 Conference*, pp. 1991–2002.
- Vignoli, M., Dosi, C. and Balboni, B. (2023) 'Design Thinking Mindset: Scale Development and Validation', *Studies in Higher Education*, 48(7), pp. 926–940.
- Spreitzer, G.M. (1995) 'Psychological Empowerment in the Workplace', *Academy of Management Journal*, 38(5), pp. 1442–1465.
- Dweck, C.S. (2006) *Mindset: The New Psychology of Success*. New York: Random House.

Kelley's Creative Confidence

SLIBRARY 2 · SHEET 3 / 5

Four fears that block creativity

PROFESSIONAL DEFINITION

David and Tom Kelley's account of *creative confidence* identifies four fears that block adults from creative output: fear of the messy unknown, fear of being judged, fear of the first step, and fear of losing control. The Kelleys argue creativity is not a fixed trait possessed by special people but a recoverable capacity available to everyone — recoverable specifically through small structured exposures that desensitise each fear, drawing on Bandura's self-efficacy research (Kelley and Kelley, 2013).

WHO DEVELOPED IT

David Kelley (founder of IDEO and Stanford d.school) and his brother Tom Kelley (IDEO partner), in their 2013 book *Creative Confidence*, distilling thirty years of practice teaching engineers, executives and students to design.

WHY IT WAS DEVELOPED

After years of teaching at Stanford, the Kelleys observed that the bottleneck was not skill but self-belief — adults had learned somewhere that they were "not creative" and had stopped trying. The book and methodology were designed to address that learned helplessness directly.

HOW IT IS USED

Practitioners progressively expose themselves to small creative challenges — sketching, prototyping, customer interviews — building confidence through small wins, supported by coached reflection on each fear as it surfaces.

WHEN IT IS USED

Use in adult-education contexts where participants self-identify as "non-creative", in onboarding for innovation teams, and in coaching conversations where fear is blocking action.

BY WHOM IT IS USED

Coaches and L&D; professionals, MBA students approaching design thinking, technical teams asked to lead innovation, and individuals at career inflection points considering creative work.

FIGURE 1 · FOUR FEARS

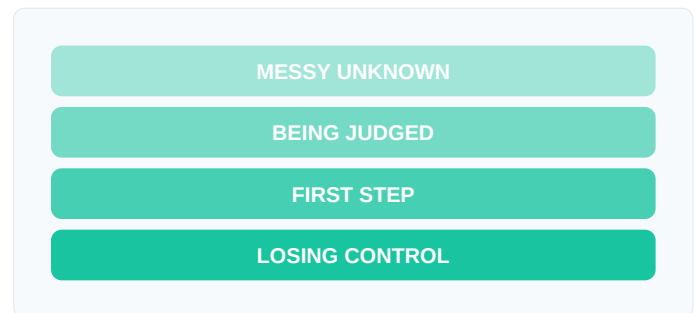


Figure 1. Each fear is addressed by a specific exposure intervention; confidence is recovered, not taught.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Robert Sternberg's (1996) creativity research identified personality traits and Mihaly Csikszentmihalyi's (1996) flow research described peak creative states, the Kelleys' contribution is a practical, fear-based recovery model rooted in Bandura's (1977) self-efficacy theory — translating creativity research into daily practice.

RELATED FRAMEWORKS IN THIS COURSE

Liedtka's Why DT Works · Slibrary 2

Dosi's DT Mindset Measure · Slibrary 2

Growth vs Fixed Mindset · Slibrary 12

Bias for Action · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

- Kelley, T. and Kelley, D. (2013) *Creative Confidence: Unleashing the Creative Potential Within Us All*. New York: Crown Business.
- Kelley, T. and Kelley, D. (2012) 'Reclaim Your Creative Confidence', *Harvard Business Review*, 90(12), pp. 115–118.
- Bandura, A. (1977) 'Self-efficacy: Toward a Unifying Theory of Behavioral Change', *Psychological Review*, 84(2), pp. 191–215.
- Csikszentmihalyi, M. (1996) *Creativity: Flow and the Psychology of Discovery and Invention*. New York: HarperCollins.

Buchanan's Wicked Problems

SLIBRARY 2 · SHEET 4 / 5

Four orders of design — signs, things, actions, systems

PROFESSIONAL DEFINITION

Richard Buchanan's seminal 1992 paper *Wicked Problems in Design Thinking* applies Rittel and Webber's (1973) concept of *wicked problems* — ill-structured, contested, with no stopping rule — to design. Buchanan organises design across four orders: signs (graphic), things (industrial), actions (interaction) and systems (environments). At each order, problems become more wicked, but the same designerly approach — deliberation through making — applies across all four (Buchanan, 1992).

WHO DEVELOPED IT

Richard Buchanan, professor at Carnegie Mellon and Case Western Reserve, in his 1992 *Design Issues* paper that became the most-cited design-thinking text in academic literature. Built on Horst Rittel and Melvin Webber's 1973 planning theory.

WHY IT WAS DEVELOPED

Design's expansion from objects to services, organisations and policy demanded a theoretical scaffold that explained why traditional problem-solving methods failed at the system level. Buchanan's four orders gave design researchers a vocabulary for that expansion and legitimised systems-level design work.

HOW IT IS USED

Practitioners diagnose which order of design they are working in. Higher orders demand more stakeholder engagement, more ethical deliberation, and longer time horizons; lower orders permit faster cycles and less consultation.

WHEN IT IS USED

Use when scoping a design engagement, when explaining to stakeholders why a project is harder than it looks, and when transitioning from product design to service or policy design.

BY WHOM IT IS USED

Design educators and researchers, service designers, public-sector innovation teams, and senior leaders sponsoring transformation programmes that span multiple orders.

FIGURE 1 · FOUR ORDERS OF DESIGN

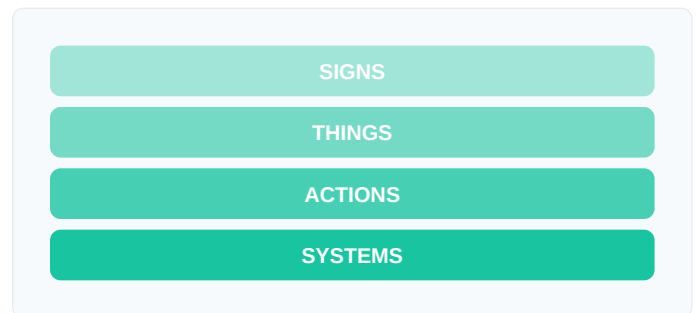


Figure 1. Wickedness intensifies with each order; design method must scale accordingly.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Rittel and Webber (1973) identified wicked problems abstractly and Herbert Simon (1969) framed design as bounded-rationality search, Buchanan's contribution is the four-orders typology — making the concept operational in design practice and giving practitioners a way to read which order of complexity they are in.

RELATED FRAMEWORKS IN THIS COURSE

Tim Brown's DT Triple · Slibrary 1

Bason & Austin: Right Way to Lead DT · Slibrary 2

12 Leverage Points · Slibrary 26

Iceberg · 4 Levels · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Buchanan, R. (1992) 'Wicked Problems in Design Thinking', *Design Issues*, 8(2), pp. 5–21.

Rittel, H.W.J. and Webber, M.M. (1973) 'Dilemmas in a General Theory of Planning', *Policy Sciences*, 4(2), pp. 155–169.

Simon, H.A. (1969) *The Sciences of the Artificial*. Cambridge, MA: MIT Press.

Cross, N. (2011) *Design Thinking: Understanding How Designers Think and Work*. Oxford: Berg.

Bason & Austin: Right Way to Lead DT

SLIBRARY 2 · SHEET 5 / 5

Six tensions leaders must navigate to lead DT well

PROFESSIONAL DEFINITION

Christian Bason and Robert Austin's research, drawn from interviews with senior leaders of design-driven organisations, identifies six tensions that determine whether design thinking succeeds or fails as a leadership practice: certainty vs uncertainty, control vs autonomy, focus vs exploration, expertise vs naivety, individual vs team contribution, and finished vs iterative. Effective leaders move *between* the poles deliberately, sequencing each side at the right moment rather than collapsing to one (Bason and Austin, 2019).

WHO DEVELOPED IT

Christian Bason, CEO of the Danish Design Centre, and Robert D. Austin, professor at Ivey Business School, in their 2019 *Harvard Business Review* article based on multi-year case studies of senior design leaders across public and private sectors.

WHY IT WAS DEVELOPED

Most design-thinking literature focuses on practitioners. Bason and Austin observed that the limiting factor at scale was leadership behaviour — leaders trained to deliver certainty struggled to host uncertainty productively. Their six-tension model gave leaders a diagnostic vocabulary for their own practice.

HOW IT IS USED

Leaders self-assess which pole they default to under pressure, then design rituals that activate the under-used pole — for example, scheduling exploration time when their bias is toward focus.

WHEN IT IS USED

Use when senior teams adopt design thinking, when DT initiatives stall under leadership pressure, in 360-feedback processes for innovation leaders, and when designing leadership-development curricula for innovation roles.

BY WHOM IT IS USED

C-suite leaders sponsoring DT, innovation directors managing portfolios, executive coaches, and HR business partners designing leadership programmes for innovation contexts.

FIGURE 1 · SIX LEADERSHIP TENSIONS

Default pole	DT pole
Certainty	Uncertainty
Control	Autonomy
Focus	Exploration
Expertise	Naivety
Finished	Iterative

Figure 1. Effective leaders move between poles deliberately rather than collapsing to one. Adapted from Bason and Austin (2019).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Roger Martin (2009) argued for integrative thinking as a leadership trait and Tim Brown (2008) emphasised the team's mindset, Bason and Austin's contribution is a leader-specific diagnostic. Building on Edmondson's (2012) team-leadership work, the six-tension model translates design ambidexterity into measurable leadership behaviour.

RELATED FRAMEWORKS IN THIS COURSE

Liedtka's Why DT Works · Slibrary 2

Buchanan's Wicked Problems · Slibrary 2

Ambidextrous Organisation · Slibrary 20

Causal vs Effectual · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

Bason, C. and Austin, R.D. (2019) 'The Right Way to Lead Design Thinking', *Harvard Business Review*, 97(2), pp. 82–91.

Bason, C. (2017) *Leading Public Design: Discovering Human-Centred Governance*. Bristol: Policy Press.

Edmondson, A.C. (2012) *Teaming: How Organizations Learn, Innovate, and Compete in the Knowledge Economy*. San Francisco: Jossey-Bass.

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Empathy Map

SLIBRARY 3 · SHEET 1 / 5

Says · Thinks · Does · Feels — the canonical user canvas

PROFESSIONAL DEFINITION

The Empathy Map is a four- or six-quadrant canvas (Says · Thinks · Does · Feels — sometimes plus Sees · Hears · Pains · Gains) used to consolidate qualitative research about a single user into one shared artefact. By separating observable behaviour (Says/Does) from inferred internal state (Thinks/Feels), the map forces the team to distinguish data from interpretation — and to notice contradictions between what users say and what they actually do (Gray, 2017; Osterwalder et al., 2014).

WHO DEVELOPED IT

Originally developed by Dave Gray at XPLANE in the early 2000s, popularised through Gray, Brown and Macanujo's *Gamestorming* (2010), and integrated into the Stanford d.school's tool stack and Osterwalder's *Value Proposition Design* (2014).

WHY IT WAS DEVELOPED

Field-research notes were unwieldy and individual to each researcher; teams could not act on them collectively. Gray needed a single visual artefact that compressed a user's world onto one wall — and that exposed the assumptions hiding inside research summaries.

HOW IT IS USED

After interviewing or observing users, the team populates each quadrant with sticky notes — colour-coded by source — for one persona. Contradictions between Says and Does become the most generative discussion points.

WHEN IT IS USED

Use after empathy interviews, before defining problems, during persona development, and as a consolidation step at the end of any qualitative research phase.

BY WHOM IT IS USED

UX researchers, product managers, service designers, and any team that has done customer interviews and needs to convert raw notes into shared insight.

FIGURE 1 · FOUR QUADRANTS

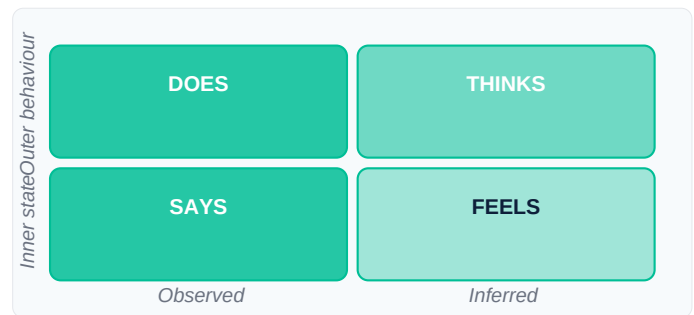


Figure 1. Says/Does are observable; Thinks/Feels are inferred. Contradictions across the diagonal are the richest insights.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Cooper's (1999) personas captured user attributes and Beyer and Holtzblatt's (1998) contextual inquiry captured behaviour, the Empathy Map's contribution is the explicit separation of observed from inferred — the design move that forces empathy to be evidence-based, not projected.

RELATED FRAMEWORKS IN THIS COURSE

[Persona Canvas · Slibrary 3](#)
[Customer Journey Map · Slibrary 3](#)
[Stanford d.school 5-Stage Process · Slibrary 1](#)
[JTBD Canvas · Slibrary 13](#)

SOURCES (HARVARD REFERENCING STYLE)

Gray, D. (2017) *Updated Empathy Map Canvas*. Available at: medium.com/the-xplane-collection.

Gray, D., Brown, S. and Macanujo, J. (2010) *Gamestorming*. Sebastopol, CA: O'Reilly.

Osterwalder, A., Pigneur, Y., Bernarda, G. and Smith, A. (2014) *Value Proposition Design*. Hoboken, NJ: Wiley.

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Persona Canvas

SLIBRARY 3 · SHEET 2 / 5

Archetype + goals + frustrations + context on one sheet

PROFESSIONAL DEFINITION

The Persona Canvas compresses the most actionable findings about a user segment into a single sheet — a fictional but evidence-based archetype with name, photo, demographics, behavioural patterns, primary goals, key frustrations, and decision context. Personas serve as proxies in design conversations, allowing teams to make user-centred trade-offs even when no real user is in the room. They are most powerful when grounded in field research, not stakeholder assumption (Cooper, 1999; Pruitt and Adlin, 2006).

WHO DEVELOPED IT

Alan Cooper introduced the persona concept in *The Inmates Are Running the Asylum* (1999), drawing on his interaction-design practice. John Pruitt and Tamara Adlin's *The Persona Lifecycle* (2006) operationalised it as a managed organisational artefact, and the canvas form was popularised by the d.school.

WHY IT WAS DEVELOPED

Cooper's clients designed for "the user" — an abstraction that meant whatever the loudest stakeholder wanted. Personas forced design conversations to refer to a specific, evidence-based archetype, displacing internal politics with external grounding.

HOW IT IS USED

Build personas from field research data, not stakeholder workshops. Three to five primary personas suffice; more dilute focus. Reference personas by name in every design decision and trade-off conversation.

WHEN IT IS USED

Use after a wave of user research, when scoping a product or service, when evaluating feature requests, and when onboarding new team members to the user context.

BY WHOM IT IS USED

UX researchers, product managers, service designers, and content strategists. Increasingly used by HR for employee personas and by sales for buyer personas.

FIGURE 1 · PERSONA CANVAS



Figure 1. Personas compress evidence into a memorable archetype that displaces stakeholder politics with user reality.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional market segmentation (Wendell Smith, 1956) which categorised demographically, Cooper's contribution was the *fictional individual* — a single named person whose behavioural specificity forced concrete design decisions. Pruitt and Adlin (2006) added the lifecycle discipline that prevents personas from becoming stale folklore.

RELATED FRAMEWORKS IN THIS COURSE

Empathy Map · Slibrary 3

Customer Journey Map · Slibrary 3

JTBD Canvas · Slibrary 13

Customer Segments · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

- Cooper, A. (1999) *The Inmates Are Running the Asylum: Why High-Tech Products Drive Us Crazy*. Indianapolis: Sams.
- Pruitt, J. and Adlin, T. (2006) *The Persona Lifecycle: Keeping People in Mind Throughout Product Design*. San Francisco: Morgan Kaufmann.
- Goodwin, K. (2009) *Designing for the Digital Age: How to Create Human-Centered Products and Services*. Indianapolis: Wiley.
- Smith, W.R. (1956) 'Product Differentiation and Market Segmentation', *Journal of Marketing*, 21(1), pp. 3–8.

Customer Journey Map

SLIBRARY 3 · SHEET 3 / 5

Stages × touchpoints × emotions — end-to-end experience

PROFESSIONAL DEFINITION

A Customer Journey Map plots a user's experience across phases of an engagement — typically Awareness, Consideration, Purchase, Use and Renewal — against rows showing actions, touchpoints, thoughts, emotions and pain points. The visualisation makes systemic problems visible: emotional dips that surface why users churn, touchpoint gaps that surface unmet needs, and high-friction handoffs that no single team currently owns

(Stickdorn et al., 2018; Polaine et al., 2013).

WHO DEVELOPED IT

The technique evolved through service-design practice in the 1990s and 2000s, codified in *This Is Service Design Doing* (Stickdorn et al., 2018), Polaine, Løvlie and Reason's *Service Design* (2013), and IDEO's HCD field guide (2015).

WHY IT WAS DEVELOPED

Organisations were structured around products and channels, but users experienced services across them. The journey map made the user's end-to-end view visible and gave cross-functional teams a single artefact to align around.

HOW IT IS USED

Plot stages along the x-axis, then map actions, touchpoints, thoughts, emotions and pain points as parallel rows. The emotion-curve dip is the most actionable signal.

WHEN IT IS USED

Use when redesigning a service, when investigating churn, when integrating multi-channel experiences, and when bridging product silos that share a customer.

BY WHOM IT IS USED

Service designers, customer-experience teams, marketing operations, and any team responsible for experiences that span multiple departments or channels.

FIGURE 1 · EMOTION CURVE

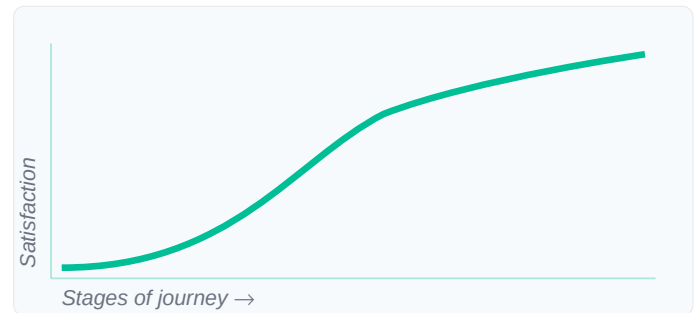


Figure 1. The dips in the emotion curve mark intervention priorities; high-emotion peaks suggest moments worth amplifying.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Shostack's (1984) service blueprint mapped operational flow and Beyer and Holtzblatt's (1998) work models documented activity, the journey map's contribution is the emotion curve — adding a measurable dimension that exposes pain points and prioritises intervention. Stickdorn's standardisation made it the lingua franca of service design.

RELATED FRAMEWORKS IN THIS COURSE

Empathy Map · Slibrary 3

Persona Canvas · Slibrary 3

JTBD Canvas · Slibrary 13

AARRR · Pirate Metrics · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

- Stickdorn, M., Hormess, M.E., Lawrence, A. and Schneider, J. (2018) *This Is Service Design Doing: Applying Service Design Thinking in the Real World*. Sebastopol, CA: O'Reilly.
- Polaine, A., Løvlie, L. and Reason, B. (2013) *Service Design: From Insight to Implementation*. Brooklyn, NY: Rosenfeld Media.
- Shostack, G.L. (1984) 'Designing Services that Deliver', *Harvard Business Review*, 62(1), pp. 133–139.
- Kalbach, J. (2020) *Mapping Experiences: A Complete Guide to Customer Alignment Through Journeys, Blueprints, and Diagrams*. 2nd edn. Sebastopol, CA: O'Reilly.

5 Whys

SLIBRARY 3 · SHEET 4 / 5

Root-cause laddering — keep asking why until you hit structure

PROFESSIONAL DEFINITION

The 5 Whys is a root-cause analysis method developed by Sakichi Toyoda at Toyota in the 1930s and codified by Taiichi Ohno (1988). Practitioners ask *why* a problem occurred, then *why* that occurred, four further times. The discipline pushes teams past the first plausible cause — typically a person — to a structural cause: a process gap, missing standard, or absent measurement that could be fixed once and prevent the whole class of failures (Ohno, 1988; Liker, 2004).

WHO DEVELOPED IT

Sakichi Toyoda, founder of Toyota Industries, and refined into the Toyota Production System by Taiichi Ohno from the 1950s. Documented in Ohno's *Toyota Production System* (1988) and popularised globally through Liker's *The Toyota Way* (2004) and Lean Startup adoption.

WHY IT WAS DEVELOPED

Manufacturing teams routinely stopped at the first cause — "the operator made an error" — and added training, only to see the same failure recur. Asking why five more times moved the diagnosis from blaming people to redesigning systems, which actually prevented recurrence.

HOW IT IS USED

When a problem occurs, ask why and document the cause. Ask why again of that cause, recursively, until you reach a structural answer (process, standard, measurement) or run out of meaningful questions — typically around the fifth iteration.

WHEN IT IS USED

Use after incidents and near-misses, in retrospectives, when investigating customer complaints, and at the synthesis stage of design research to push past surface symptoms.

BY WHOM IT IS USED

Operations and quality teams, lean practitioners, product managers diagnosing product failures, and design researchers pushing past surface complaints to underlying needs.

FIGURE 1 · WHY LADDER

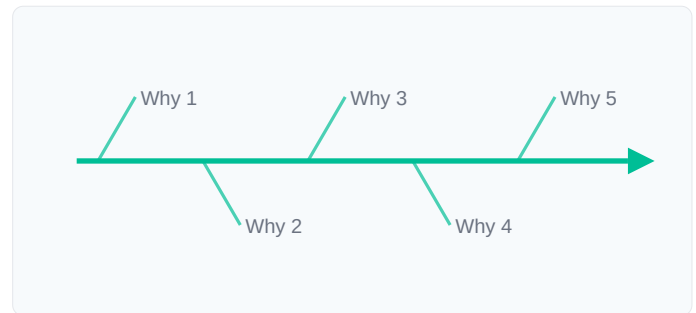


Figure 1. Each "why" descends one structural layer; five iterations typically reaches a fixable system root.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Ishikawa's (1968) fishbone diagram (which maps causes laterally) and traditional accident analysis (which stops at human error), the 5 Whys' contribution is the disciplined depth-first descent. Where Goldratt's (1984) Theory of Constraints maps systems statically, 5 Whys traces a single failure to its structural root in minutes.

RELATED FRAMEWORKS IN THIS COURSE

How Might We (HMW) · Slibrary 3

Pre-Mortem · Slibrary 11

Iceberg · 4 Levels · Slibrary 25

Riskiest Assumption Test · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

- Ohno, T. (1988) *Toyota Production System: Beyond Large-Scale Production*. Cambridge, MA: Productivity Press.
- Liker, J.K. (2004) *The Toyota Way: 14 Management Principles from the World's Greatest Manufacturer*. New York: McGraw-Hill.
- Ries, E. (2011) *The Lean Startup*. New York: Crown Business.
- Ishikawa, K. (1968) *Guide to Quality Control*. Tokyo: JUSE Press.

How Might We (HMW)

SLIBRARY 3 · SHEET 5 / 5

Turning insights into reframed, generative design questions

PROFESSIONAL DEFINITION

How Might We (HMW) is a question-stem technique that converts a user insight or problem into a generative design question. The phrasing — How (assuming a solution exists) Might (humble, exploratory) We (collaborative) — is engineered to open ideation rather than close it. A well-formed HMW is broad enough to invite diverse solutions but narrow enough to anchor them, sitting at the right altitude between problem statement and solution prompt (Berger, 2014; Liedtka and Ogilvie, 2011).

WHO DEVELOPED IT

Originated at Procter & Gamble in the 1970s under Min Basadur as part of their creative-problem-solving training, adopted and popularised by IDEO and the Stanford d.school in the 2000s, and analysed in Warren Berger's *A More Beautiful Question* (2014).

WHY IT WAS DEVELOPED

Practitioners observed that problem statements written as fact ("users abandon the form") shut down ideation, while solution prompts ("add validation") foreclosed alternatives. The HMW stem split the difference, opening the question without losing focus — and built in collaboration linguistically.

HOW IT IS USED

Take a user insight from research, then rephrase as *How might we [verb] [user] [outcome] [in context]*. Test the altitude — too broad invites everything, too narrow invites one solution. Iterate the wording before opening to ideation.

WHEN IT IS USED

Use at the bridge between empathy and ideation phases, when reframing a design brief, when challenging an assumed solution, and when team energy is flagging due to over-narrow framing.

BY WHOM IT IS USED

Design facilitators, product managers writing briefs, innovation coaches, and any team transitioning from research synthesis to ideation.

FIGURE 1 · QUESTION ALTITUDE

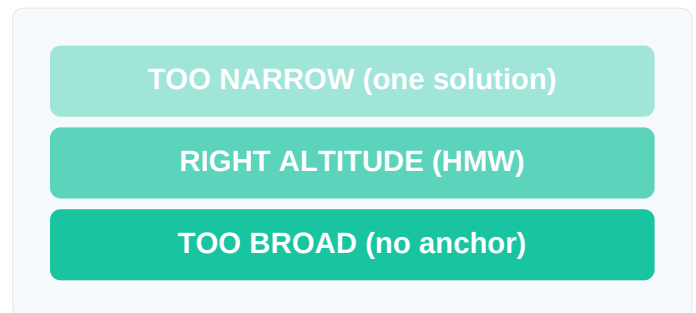


Figure 1. HMW sits between solution-prescription and undifferentiated openness. Altitude is the design.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where traditional brainstorming (Osborn, 1953) opened ideation without a question and Min Basadur's (1994) early Simplex method gave problem-finding a structure, HMW's contribution is the precise grammatical engineering of the question itself — making the question's *shape* the lever for ideation quality.

RELATED FRAMEWORKS IN THIS COURSE

5 Whys · Slibrary 3

Crazy 8s · Slibrary 4

SCAMPER · Slibrary 4

Stanford d.school 5-Stage Process · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

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